

Stateless Flight Detection on Smartphones: Confounder-Aware Evaluation and a Physical Overlap Limit

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Abstract

FLYENJOY is an iOS travel application whose in-flight social tab forms a peer-to-peer network among passengers on one aircraft. Wi-Fi and cellular are unavailable in the air, and background scheduling means the app may first run at cruise, so three production detectors built on barometric and GPS altitude failed. Each decision must therefore rest on one 30-second window with no history across launches, a case prior work does not cover: the closest system detects takeoff and landing transitions. Evaluating thirteen iOS modalities, with realistic ground confounders held out as complete folds, the barometer ranks first and sits in the lowest cost tier for power, privacy, and permissions. Cost-sensitive thresholding with a five-window debounce cuts false activations from tens per ground-day to roughly one at 450 s median latency, and 15 of 19 flight recordings activate. The remaining errors are not a coverage gap. Cruise cabin pressure is 753–838 hPa, but high-elevation ground and re-equalized cabins meet near the learned threshold: a 960–1010 hPa band holds 39% of scored windows against 84% of false positives and 74% of false negatives. Broader coverage, temporal state, and debouncing do not remove the ambiguity. A field study with the configuration frozen beforehand found the detector active for 66.9 of 67.2 ground-hours between 1,360 and 2,700 m, making high elevation a release blocker pending an abstention region or a second in-band modality. Corpus size alone cannot separate a coverage gap from an observability limit.

1 Introduction

FLYENJOY¹ is an iOS travel application with a dedicated in-flight social tab that creates a transient peer-to-peer network among passengers on the same flight. The network is deliberately flight-scoped, so the application must determine whether the device is in flight; because Wi-Fi and cellular are unavailable in the air, it cannot use connectivity to find out. The costs of detection errors are asymmetric: a false negative prevents access during a flight, whereas a false positive opens a flight-scoped social space to people who are not on the flight. An earlier version of this detector shipped as the live network gate and behaved intermittently in the field.

The detector must also run within iOS background-execution limits at a power cost low enough to operate con-

tinuously. It cannot assume that it observed takeoff, and at cold start it retains no information from earlier app launches. Every prediction therefore stands on one 30-second window of on-device sensing (Appendix C.7 varies this length). This stateless operating contract eliminates a common remedy for detection errors: the use of temporal history. Broader training coverage remains the obvious alternative. For an always-on detector, however, the negative class is effectively an open world, so whether additional data suffices is an empirical question rather than a background assumption.

Prior transportation mode detection research provides limited guidance for this setting. Standard benchmarks primarily consider walking, cycling, driving, and rail travel rather than flight. The most closely related system, flyDetect (Reinholdt, Jul, and Ferreira 2024), detects the transitions associated with takeoff and landing. A background application, however, may begin observing a flight only after the aircraft has reached cruise. Our problem therefore requires sustained state classification from an isolated sensing window rather than transition detection.

This application setting raises three questions:

- **RQ1 (capability):** Which smartphone sensing configurations support stateless flight detection under energy, permission, and availability constraints?
- **RQ2 (failure and repair):** Why does the strongest configuration fail, and can broader training coverage, temporal state, or decision logic repair it?
- **RQ3 (deployability):** Given those limits, what configuration should ship, and does it survive unseen conditions and devices?

The configuration these questions select is then deployed and measured, so the answers are tested prospectively rather than only cross-validated.

We answer these questions through an evaluation of thirteen iOS sensor modalities designed to expose realistic confounders. The corpus contains 3,737 in-flight and 5,757 ground windows, each 30 seconds long, collected from 39 recordings spanning 17 commercial flight legs and seven iPhone models. Labels are derived independently from externally sourced flight-tracking timestamps. To prevent information leakage between related windows, the evaluation holds out complete travel episodes rather than individual windows. We also collect targeted ground confounder sce-

narios that reproduce physical signals resembling those observed in flight, including interstate driving under changing ambient pressure and a high-rise environment with pressure transients. Each confounder scenario is held out as a complete evaluation fold. We then deploy the selected configuration to two participants over a five-day trip, adding 136 device-hours and two flight legs as a held-out test at a ground elevation the corpus does not sample.

The results reveal an important distinction between insufficient coverage and limited observability. The barometer is the strongest modality in the lowest deployment-cost tier, but ground and in-flight cabin pressures overlap near its learned decision boundary. High-elevation airports lower ground pressure, while cabin re-equalization during approach and low, short flights whose cabin pressure never leaves the ground range raise in-flight pressure. Although the resulting 960–1010 hPa band contains 39% of the out-of-fold scored windows, it accounts for 84% of false positives and 74% of false negatives. Broader negative class coverage, temporal state, and decision logic debouncing do not eliminate this ambiguity. Instead, they shift the balance between the two types of error. Deployment therefore requires an explicit operating point that balances false activations against flight coverage and detection latency.

This paper makes four contributions:

1. We identify, quantify, and prospectively confirm an *overlap limit* for stateless flight detection, showing that dominant failures are not merely missing negative coverage and that the tested remedies cannot eliminate the two-sided ambiguity.
2. We introduce a confounder-aware protocol for always-on detectors that holds realistic confounders out as complete folds, reports them separately, and tests whether errors survive deployment decision logic.
3. To our knowledge, we provide the first systematic comparison of thirteen iOS modalities for single-window flight detection across accuracy, availability, model capacity, power/privacy cost, permissions, device transfer, and deployment constraints.
4. We report a targeted operational field stress test with model, threshold, and debounce fixed before collection. It confirms corpus predictions, supplies missing high-elevation ground, and exposes premature release and sampling-rate dependence.

2 Deployment Setting and Prior Attempts

Before this study, we built and tested three flight detectors, and none worked well enough. Their failures exposed the constraints that define the deployment setting studied here.

Attempt 1: barometric altitude. The first detector used relative barometric altitude to detect a climb. It fails when the app starts mid-flight because the baseline is established at launch, so cruise contains no observed ascent. Switching to absolute altitude did not solve the problem.

Attempt 2: GPS altitude and speed. The replacement classified a location fix using altitude and ground speed and was deployed as the network gate. Its behavior was intermittent

for a reason visible in the corpus: Location is simply unavailable for much of a flight. Under the intended radio configuration it reads on only 36.1% of in-flight windows against 72.6% of ground windows (Table 7). A detector gated on GPS is gated on a signal that is absent precisely when it is needed, and airplane mode removes or restricts several of the sources Core Location can draw on.

Attempt 3: averaged GPS altitude. We initially treated this as a measurement-noise problem and responded with duty-cycled GPS sampling and averaging. The deeper issue was availability rather than variance: obtaining enough readings required loosening the vertical-accuracy criterion from 50 m to 200 m. Averaging can reduce noise in an observed signal; it cannot recover a signal that is frequently absent.

Acceleration-based takeoff triggers were excluded because background execution cannot guarantee observation near wheels-up; engine-noise detection requires microphone permission and continuous high-power sampling.

Reframing the problem. The failed approaches largely ask *how high above sea level is this device?* In this setting that quantity is either unavailable or misleading. GPS is unreliable inside the aircraft, while cabin pressurization deliberately decouples interior pressure from outside altitude. Cruise cabin pressure sits near 750–840 hPa (753–838 across this corpus), whereas a 10,000 ft altitude threshold corresponds to about 697 hPa. The useful quantity is therefore not inferred altitude but *absolute cabin pressure*, which requires no satellite fix, no network, and no launch-time baseline. It is not permission-free (CMAltimeter is gated on the motion authorization and its usage description), but that prompt is lower-friction and less privacy-bearing than the separate Location and microphone prompts. Pressurization makes this signal informative because such pressures are uncommon on the ground, although Section 6.2 shows why uncommon is not impossible and Section 6.4 measures how often: 66.9 of 67.2 ground-hours at 2,000–2,700 m.

Deployment contract. These failures define *stateless features with bounded, fail-safe decision memory*. Features use only their own window, never earlier windows, recording history, or a persistent baseline. The only memory is a debounce counter that resets across launches and defaults to no activation. We use absolute pressure and within-window step deltas rather than persistent baselines or counts.

Because false activations open the ground network, we select the operating point explicitly and report event-level false activations and detection latency alongside window accuracy. Latency is measured both from wheels-up and from the detector’s first observed window, separating passenger wait from background scheduling.

3 Related Work

Automatic flight detection has used acoustics, accelerometers, and smartphone sensors. Tawk et al. (Tawk et al. 2013) gate a package-tracking radio using static-versus-dynamic acceleration, without separating aircraft from cars. Barille (Barille 2024) measures cabin pressure across flight phases with a smartphone barometer but builds no classifier: it establishes the signal we exploit, though not the

ground conditions that overlap it. The closest system, flyDetect (Reinholdt, Jul, and Ferreira 2024), detects takeoff and landing transitions; our background detector may first run at cruise and must classify a single window with no prior state.

The broader transportation-mode-detection literature provides mature methods and datasets, including the Sussex-Huawei Locomotion corpus (Gjoreski et al. 2018; Wang et al. 2019) and GPS-trajectory datasets such as GeoLife (Zheng, Xie, and Ma 2010). The former lacks an airplane class, while the latter contains airplane-labeled GPS trajectories without complementary onboard sensor data. Both are based on temporal or continuous observations rather than the isolated-window setting required by our deployment contract. Travel-diary and trip-annotation systems (Montini et al. 2014) likewise analyze journeys post hoc rather than detecting flight on-device in real time. Our setting therefore differs in both scope and constraint: thirteen sensing modalities, one 30-second window, and no historical state.

4 Corpus and Evaluation Design

4.1 Confounder-aware evaluation

For an always-on detector, the negative class is open world. We therefore evaluate ordinary generalization and mechanism-predicted failures using four principles:

- I **Confounder scenarios.** Collect realistic negatives that reproduce the strongest signal’s physical mechanism without the target event.
- II **Whole-fold holdout.** Hold each scenario out completely to test first-encounter generalization.
- III **Per-fold reporting.** Report each fold separately so catastrophic failures are not hidden by pooling.
- IV **Decision-level survivability.** Test whether clustered errors survive deployment logic such as debouncing.

The protocol is mechanistic, not exhaustive: finite data cannot guarantee an open-world negative class. Appendix E.2 generalizes the procedure.

4.2 Corpus

We classify non-overlapping 30-second windows from sustained ground and flight periods, excluding transitions. The corpus has 9,494 labeled windows, 3,737 flight and 5,757 ground, from 39 usable recordings spanning 17 flight legs, eight ground scenarios, and seven iPhone models. Of 43 recordings collected, four are labeled transitional and contribute sampling rates only; the remaining 39 yield the labeled windows, 79.1 of the 79.9 recorded hours. Evaluation folds are per trip; deployment metrics are per flight recording.

The 19 recordings with wheels-up references cover 16 of 17 flight legs. Four legs were recorded by multiple devices, and two recordings each span both legs of a connection. Each is a continuous SensorLogger session with one file per sensor, collected under natural use; all flight recordings used airplane mode. Labels come from FlightRadar24 or FlightAware. Neither exported altitude field is height above

ground, so a zero reading is not in general a ground contact; in these exports, however, ground rows carry a literal 0 sentinel alongside taxi-speed ground tracks, while the first airborne sample appears at pressure altitude. We therefore take wheels-up and touchdown as the first and last non-zero altitude point ± 30 s. Because sensor coverage varies, every per-sensor result reports fold support. Sample-rate estimates use all 43 recordings for motion sensors and barometer, but 17 for microphone and pedometer.

Pressure characteristics. Cruise cabin pressure ranges from 753–838 hPa (Table 4) and spans 85 hPa across trips, roughly 2,800 ft of cabin altitude. Because cabin pressure is set by a pressurization schedule rather than by outside altitude, it is not recoverable from cruise altitude. We do not record aircraft type or cruise altitude per trip and make no claim about which aircraft types sit where in this range. Ground windows span 970.9–1026.7 hPa, a floor set by where this corpus was collected rather than by ground pressure generally (Section 6.4 observes ground windows to 734 hPa); in-flight windows span 752.6–1027.3 hPa, with high values during approach re-equalization or low sectors that never leave ground range. The overlap motivates RQ2.

Confounder scenarios. Five ground-only recordings capture two flight-like barometric scenarios. *High-rise* has 429 windows across two sessions, including 53 minutes on a 23rd floor and six elevator round trips; pressure remains above 993 hPa, isolating within-window change. *Interstate driving* has 637 windows over five hours, spanning 979.6–1000.5 hPa. Both are held-out deployment conditions.

4.3 Protocols, folds, and scoring

We evaluate three holdout protocols. **P1** holds out a trip, **P2** a ground scenario, and **P3** both: for each of $11 \times 8 = 88$ trip-scenario pairs, the full trip is tested while that scenario is removed from training. Thus P3 is a training ablation on an unseen flight, not a test of the held-out ground condition; P2 scores that condition. P3 is primary. For each sensor, P3 averages normalized AP over every *supported* trip $\times$ scenario cell: 88 where a sensor reads on all eleven trips, 80 for Location and 30 for the microphone. We average cells rather than pooling windows because every trip appears once per scenario, so pooling would count each window up to eight times; we therefore report no pooled AP for P3, and the *folds* column of Table 1 is the P1 fold count. Every fold holds out a complete travel episode, including connected legs and devices recording the same episode, preventing train-test leakage. Eleven folds contain both classes. P2 folds are ground-only, so we report false-positive rate rather than AP.

Normalized average precision. Because AP’s no-skill value is positive prevalence, which ranges from 11.3% to 99.1% across folds, we report $AP_{\text{norm}} = \frac{AP - \pi}{1 - \pi}$, where π is fold prevalence. Values 0, 1, and below 0 denote no skill, perfection, and below-baseline performance. The normalization divides by $1 - \pi$, so folds near $\pi = 1$ magnify small AP changes: T30 is 99.1% positive (5 ground windows against 579 flight), where the factor is roughly 117. Fold-mean normalized AP should therefore be read as a ranking statistic, and the intervals we report bracket pooled AP rather than the fold mean.

Table 1: Per-sensor detection under leave-one-trip-out, sorted by normalized AP. The interval is a trip-level cluster bootstrap and brackets *pooled* AP (not normalized column).

Sensor	norm. AP		pooled AP		folds
	P1	P3	value	95% CI	
Barometer	0.894	0.885	0.958	[0.895, 0.993]	11
Microphone	0.794	0.766	0.911	[0.864, 0.972]	5
Location	0.743	0.737	0.960	[0.750, 0.992]	10
Accelerometer	0.456	0.455	0.753	[0.649, 0.852]	11
Gyroscope	0.339	0.354	0.680	[0.540, 0.799]	11
Orientation	0.220	0.231	0.561	[0.444, 0.706]	11
Magnetometer	0.220	0.218	0.584	[0.416, 0.782]	11
Battery	0.208	0.202	0.663	[0.439, 0.836]	8
Compass	0.205	0.201	0.507	[0.335, 0.713]	11
Network	0.119	0.119	0.622	[0.316, 0.863]	10
Gravity	0.067	0.070	0.495	[0.369, 0.660]	11
Brightness	0.037	0.018	0.537	[0.219, 0.869]	8
Pedometer	0.033	0.033	0.425	[0.268, 0.699]	11

Intervals and resolution. Out-of-fold scoring covers 8,363 of 9,494 labeled windows, including 4,691 ground windows. Band shares, error shares, and event rates use this scored set, so false activations are per 1.63 ground-days of scored exposure. Confidence intervals resample trips. The barometer resolution floor, half the trip-level interval width, is ± 0.049 ; smaller differences are unresolved. Appendix E details folds and resampling.

5 Models and Decisions

Windowing and features. Readings form non-overlapping 30-second windows using within-window means, standard deviations, ranges, magnitudes, and related statistics (Appendix A.3); features never cross window boundaries, and sensors are modeled independently.

Classifiers and controls. For each sensor, we train a depth-4 Decision Tree with balanced class weights and Logistic Regression in scikit-learn (Pedregosa et al. 2011). The tree is primary; logistic regression is a structural control because a linear boundary cannot directly isolate a bounded pressure interval. Claims are tested under both on identical folds, with disagreements labeled model-dependent. Random forest and gradient boosting provide capacity ceilings (Appendix E.5), not controls. Commodity classifiers keep the analysis focused on sensor information rather than model novelty.

Stateful variants. For RQ2, diagnostic variants add drift from recording start, local pressure trend, and a trailing rolling mean. These intentionally violate cold-start contract.

6 Results

Unless noted, results use Section 4.3’s protocols, stateless 30-second windows, and trip-level intervals.

6.1 RQ1: What sensing configuration supports stateless flight detection?

Per-sensor detection accuracy. Across eleven both-class folds (Table 1), barometer leads at normalized AP 0.894 un-

der P1 and 0.885 under primary P3, with pooled AP 0.958 [0.895, 0.993]. Microphone follows at 0.794 on fewer trips, then Location at 0.743; motion sensors are substantially lower. Pooled AP makes barometer and Location nearly equal, 0.958 versus 0.960; normalizing those by pooled prevalence (0.439 versus 0.280) gives 0.926 and 0.944 (Table 7), whereas fold-mean normalized AP is 0.894 versus 0.743. Tenth-percentile fold scores are 0.760 and 0.287, respectively: strong Location folds mask GPS-absent ones when pooled. We therefore lead with fold means because deployment must survive individual trips.

The ranking is not an artifact of the model. Barometer, microphone, Location, and accelerometer remain the top four, in order, under the tree, logistic regression, random forest, and gradient boosting. Extra capacity improves barometer pooled AP by only +0.015 [−0.004, +0.050] on a paired trip-level bootstrap, so the gain is unresolved. Below rank four, ordering is capacity-bound, so we interpret only the top four (Figure 1; Appendix E.5).

Which configurations survive hard negatives. Each purpose-built ground confounder is held out completely; because these folds have undefined AP, Figure 2 and Table 3 report false-positive rate. Magnetometer and compass fail indoors; magnetometer reaches 0.72 on high-rise. Gyroscope reverses the pattern, 0.31 indoors but 0.96 in a car. These patterns are consistent with, but do not experimentally isolate, altered indoor magnetic fields and low angular motion in a phone mount. *Only barometer and Location stay low on both* (worst-case 0.14 and 0.24). Flight-only scoring could therefore select a gyroscope model that labels 96% of a highway drive as flight. Both confounders are needed to expose the pattern.

Fusion and barometer-free devices. A nested-CV logistic stacker over all sensors reaches pooled AP 0.984 [0.954, 0.998] versus 0.958 [0.895, 0.993] for barometer; the paired trip-level gain is +0.026 [+0.004, +0.063]. Without barometer, pooled AP is 0.926 [0.856, 0.977], an unresolved −0.032 [−0.078, +0.010]. Greedy selection falls from 0.939 to 0.901 when each held-out trip is excluded from selection, revealing +0.038 selection bias (Appendix C.2).

Availability is part of accuracy. Observed-only scores ignore silence. On the barometer’s full out-of-fold grid, assigning $p = 0$ when a sensor is absent, Location drops from 0.944 to 0.336 normalized AP and microphone from 0.868 to 0.261; barometer remains 0.926 (Figure 3). Location reads on 72.6% of ground but only 36.1% of flight windows, reproducing Attempt 2.

Cost. Barometer dominates the deployment dimensions we characterize: highest normalized AP, lowest expected power and privacy cost, availability whenever the app executes, 0.9 Hz sampling, and only motion authorization rather than a separate privacy-bearing prompt (Table 9). GPS and microphone add permission gates, high power, and background restrictions. A composite score is therefore unnecessary; Appendix C.7 reports one only illustratively. *The selected classifier uses barometer features alone*; accelerometer and pedometer are near-zero-marginal-cost coprocessor peers, not model inputs.

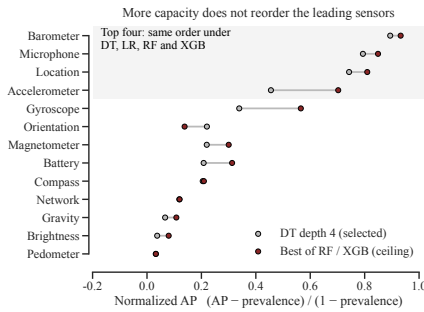


Figure 1: Depth-4 tree against the best of random forest and gradient boosting, in normalized AP. The barometer gains +0.038 here, +0.015 in pooled AP.

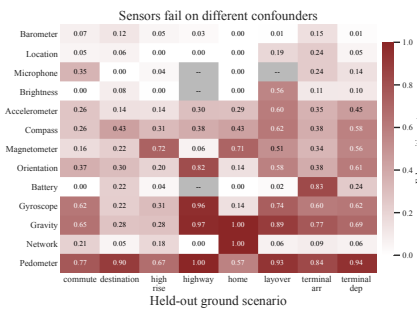


Figure 2: False-positive rate per (sensor, scenario) under the P2 hold-out, sorted by worst case. Only the barometer and Location have no dark cell.

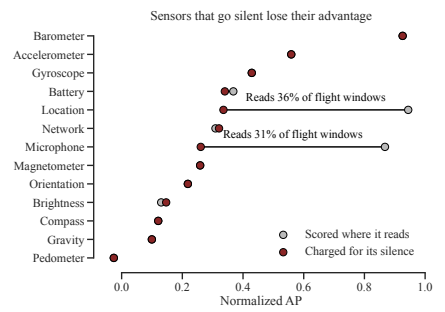


Figure 3: Normalized AP scored where a sensor reads against scored on the whole decision grid, charging silence as a fail-safe. Barometer’s two points coincide.

6.2 RQ2: Why does the best detector fail, and can the failure be repaired?

What the barometer learns, and the band where it cannot. On the full dataset, the tree classifies `pressure_mean` ≤ 970.88 hPa as flight; above that split, flight requires large `pressure_range`. Cruise cabin pressure is 753–838 hPa, so pressure level supplies the dominant signal. Absolute pressure still overlaps (Figure 4). Elevated airports lower ground pressure, while approach re-equalization and low sectors raise flight pressure. Although all 5,757 ground windows lie above the full-data split, so do 630 of 3,737 flight windows (16.9%). Among 72 candidate bands on a fixed 5 hPa grid fixed before scoring, we report the one spanning most of both error types rather than the most concentrated: 960–1010 hPa covers 39.3% of the out-of-fold scored windows yet contains 84.5% of false positives and 74.4% of false negatives; both error directions occupy the same physical region.

Repair 1: negative-class coverage. Adding in-band ground negatives shifts the threshold. Withholding the in-flight half of the band from training, with code, folds, and features fixed, lowers ground false-positive rate from 0.0770 to 0.0281 and raises pooled AP from 0.9584 to 0.9619. In isolation, this looks like a repair. The manipulation is itself unresolved and model-dependent: logistic regression never exceeds 0.0003 and none of twelve capacity $\times$ weighting intervals excludes zero (Appendix E.5). We therefore rest the argument on where the errors sit, not on the size of this shift. The detector then loses the withheld flight population: recall is 0.158 [0.107, 0.259] on approach-equalized windows versus 0.975 [0.955, 0.987] on ordinary flight, and 0.057 on two low-altitude-cruise recordings. The 279-window ground-pressure zone across 15 recordings has negative normalized AP under both models: -0.857 [$-1.325, -0.133$] for the tree and -0.956 [$-1.547, -0.245$] for logistic regression. Moving the threshold therefore exchanges false positives for missed flight; no threshold makes both small (Figure 5; Appendix B.1).

Repair 2: temporal state. Adding history does not consistently improve the barometer. Δ_{state} is -0.0062 [$-0.0151, +0.0015$] for the tree and $+0.0043$

[$-0.0008, +0.0116$] for logistic regression; the sign changes and neither interval excludes zero. Both effects are below 0.007 in magnitude, with intervals extending to about 0.015. Below eight training trips, random subsets straddle zero. More fundamentally, history cannot rescue a *cold start* that begins inside an already equalized cabin, which is the case the contract must survive. A detector that had observed the ascent could carry flight state through approach; Section 6.4 measures how much a trajectory would buy, at the cost of the stateless contract.

Repair 3: decision-level debouncing. Requiring k consecutive positives removes isolated errors, but overlap errors cluster. Increasing k from 1 to 5 changes false-positive windows from 222 to 196 per day while distinct activations fall from 33.8 to 6.1 at the untuned threshold. Debouncing improves decisions without resolving overlap.

The limit. None of the tested remedies closes the overlap. Broader coverage trades one error direction for the other, temporal-state effects are unresolved and model-dependent, and debouncing suppresses repeated decisions rather than ambiguity. The failure is therefore an observability limit of the tested one-window pressure representations, shown across four model classes and a richer representation in Appendix E.6; we do not claim an information-theoretic limit over raw pressure. Two checks favor a physical explanation over label noise. Pressure-defined hard windows separate much better than transition-proximal ones: the ground-pressure zone falls to -0.857 normalized AP while windows within four minutes of a transition fall only to 0.322 (Appendix B.1). Also, although cabin pressure takes a median 5.3 minutes after departure to cross the split (Figure 6), approach-equalized windows occur at flight end, where labels are unambiguous.

What found this, and what did not. The confounder protocol alone could not reveal two-sided overlap because its extra recordings are ground-only. They provide one quarter of the band’s ground side and none of its flight side; 37% of the flight side comes from later data. Hardening the negative class was necessary but not sufficient. The confounder recordings helped expose the ground side of the failure, but the two-sided overlap became clear only after later in-flight

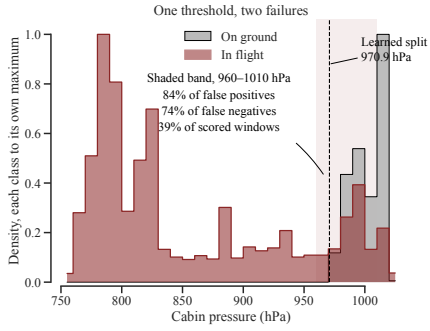


Figure 4: Cabin pressure, ground against flight, each class scaled to its own maximum. Dashed line is the learned split; shaded band is 39% of scored windows.

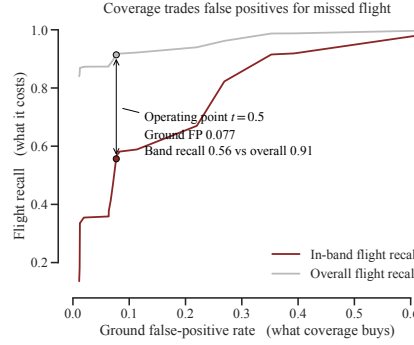


Figure 5: Ground false-positive rate against flight recall as the decision boundary moves. The two error directions trade against each other.

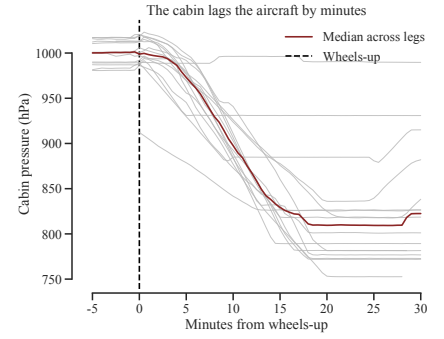


Figure 6: Cabin pressure aligned to wheels-up, one trace per flight, with the median overlaid. Median time to cross the split is 5.3 minutes.

recordings populated the same pressure range. Additional regional turboprop and approach recordings provided that missing positive-class evidence.

6.3 RQ3: Given that limit, what should ship?

Per-window error rate does not predict deployability. Barometer false-positive rates are similar on high-rise (0.047) and driving (0.027), but run lengths differ. No driving error exceeds four windows, so none survives $k=5$; lift errors reach eight windows, and one third survive. Still longer runs occur at elevated airports: 51 consecutive windows (25 minutes) on arrival and 176 (88 minutes) during a destination stay. Those windows lie at 970.9–977.0 hPa, above the full-data split, so the run reflects that fold’s threshold rather than either branch of the quoted full-data rule. The high-rise data separate transient from sustained errors. Under the full-data fit, all 26 false-positive windows occur during lift transients, firing at 0.769. Table 3’s 0.047 instead scores the P2 model trained without high-rise. The other 403 high-rise windows, including a 53-minute plateau at 999 hPa, 28.5 hPa above the split, produce zero false positives. Lift errors are thus bounded by the ride, although runs up to eight windows can survive $k=5$. Elevation errors are sustained and survive any finite debounce while the device remains elevated. The two scenarios differ by only 0.020 in false-positive rate but have different deployment consequences.

The failures are partly model-dependent. Across sensor-scenario false-positive rates, median per-sensor tree/logistic correlation is 0.437, and 13 of 54 cells have non-overlapping intervals. We therefore treat failure direction as robust, not every magnitude.

False activations and latency. At threshold 0.5, barometer produces 33.8 false activations per day and detects 16 of 19 flight recordings at 30 s median latency from wheels-up. This does not conflict with the 5.3-minute median pressure-mean crossing: under the full-data diagnostic fit, the rule’s pressure-range clause fires during climb, recovering 291 of 630 in-flight windows above the split at the cost of 96 false positives among 5,757 ground windows, both full-corpus de-

nominators. Raising the threshold to the cost-selected $\theta = 0.843$ cuts activations to 4.3 per day; adding $k = 5$ reduces them to 0.6 at 450 s median latency. Across all ground data, $\theta = 0.843$ yields seven distinct activations without debounce and one with it. We choose $k = 5$ for margin, not because the corpus resolves an advantage over $k = 3$. Both θ and k are selected on the same out-of-fold predictions used for scoring. Nested policy selection roughly triples the rate, but event counts are too small to resolve the difference (Appendix C.6). Logistic regression shows the same trade (19.0 $\rightarrow$ 1.2 per day). False-positive windows change only from 222 to 196 per day, showing why event-level metrics matter.

Latency has two tails. From wheels-up, P90 is 7,140 s and the maximum 9,060 s, but four of 19 recordings begin 14.5–129 minutes after wheels-up because background scheduling controls observation time. Three of those four fire within 120 s of their first window, the floor at $k = 5$ under our timestamp convention. From first observation, median latency is 330 s, P90 984 s, and only one recording remains slow at 150 minutes. Most of the wheels-up tail is therefore scheduling, not sensing. The cumulative curve (Figure 7) gives its shape: ten of 19 recordings activate within eight minutes of wheels-up, five between 16.5 and 151 minutes, and four never. The product therefore needs a fallback for slow activation rather than a tighter latency target the signal cannot support.

The overlap appears in the other tail as failure rather than delay. Undetected recordings have 0.46 of in-flight windows above the learned split versus 0.18 for detected recordings (Mann–Whitney $p = 0.04$); one never leaves ground pressure. Among detected recordings, latency is unrelated to that fraction ($\rho = 0.16$, $p = 0.58$). Thus the overlap mainly causes missed activation rather than simply slower detection, while most of the long wheels-up latency tail comes from delayed background observation. We therefore report coverage and latency together. Both statistics rest on four undetected recordings and are substantially driven by the one fully in-band case.

Calibration and operating point. Barometer probabilities

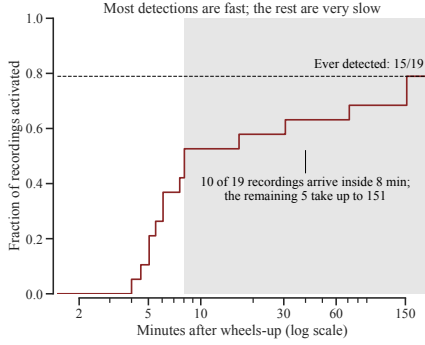


Figure 7: Fraction of flight recordings activated within m minutes of wheels-up, log time. Ten arrive inside eight minutes; five straggle out to 151.

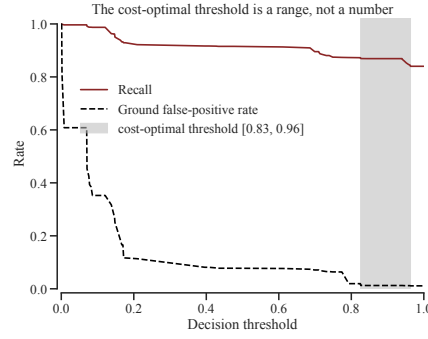


Figure 8: Recall and ground false-positive rate against the decision threshold. The shaded band is the cost-optimal threshold under a 10:1 cost.

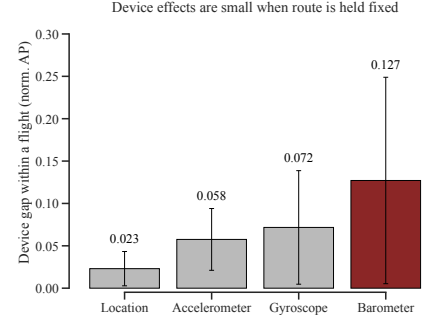


Figure 9: Matched within-flight cross-device gap: the same flight on two devices at once, holding route, weather and aircraft fixed.

are reasonably calibrated (Appendix C.4), and a 10:1 false-positive cost shifts the operating point away from threshold 0.5 (Figure 8). Because trip-level spread is wide, we report a cost-optimal range; the sweep optimizes window recall and false-positive rate, not post-debounce events, so it identifies a defensible region rather than an event-optimal point.

Cross-device transfer. Simultaneous recordings on different devices partially separate route from device effects. Across four shared sensors, the average device gap is 0.070; barometer is 0.127 (Figure 9). Across three pairings, iPhone 16 reads 0.50–0.55 hPa above iPhone 13 Pro Max, with within-pair standard deviation below 0.09, so an absolute-pressure rule inherits device bias (Appendix C.3).

The implemented configuration. The selected policy is barometer-only with three within-window features, 30 s windows, a depth-4 tree at $\theta = 0.843$, symmetric $k_{\text{on}} = k_{\text{off}} = 5$, and fail-safe closed cold start (Appendix Table 8).

The fail-safe gate stays closed when a window cannot be resolved, so overlap primarily costs coverage. Section 6.4 tests this configuration outside corpus conditions.

6.4 Post-deployment field study

Ordinary deployment telemetry cannot close the gaps the corpus leaves. Scoring the detector requires knowing whether the device is airborne, and establishing that continuously would mean collecting the persistent location or travel state the detector exists to avoid needing, so a population accuracy figure is not missing data but a boundary the product sets. We therefore ran a consented, instrumented *operational stress test* of the fixed configuration, designed to exercise three blind spots this paper has already named: high-elevation ground (Section 6.2), a device and OS absent from the corpus (Section 6.3, where leave-one-device-out is confounded by route), and the pair-level concurrency the peer network actually requires but no single-device metric expresses. Two participants carried the configuration through a five-day Seattle–Bozeman round trip with three days at Yellowstone, 2,000–2,700 m above sea level. The study adds 136.3 device-hours on two iPhone generations and two flight legs, with flight-tracker wheels-up/down references. It diag-

noses mechanisms and severity, not population rates.

This is a held-out test. The tree, $\theta = 0.843$ and $k=5$ were fixed on the corpus before collection, and no field window was used for fitting or threshold selection. The one post-hoc change is an implementation detail: the shipped sample-count guard is an absolute count where it should scale with sampling rate (Appendix D).

Table 2: The corpus’s predictions against the deployment. Rows 1–4 test figures fixed before collection; the last two have no offline counterpart, because window-level scoring cannot express them.

	Predicted (corpus)	Measured (field)
Elevated gate	flight, <i>qualitative</i>	120/120 windows, 862–868 hPa
Activation latency	450 s median	414 s, both devices
Cabin crosses split	5.3 min median	6.4 min
False activations, low elevation	0.6 per ground-day	0 in 2.65 ground-days
Release vs wheels-down	—	–4.7 min, both devices
Per-session sampling rate	—	0.791–1.089 Hz, one device

The predicted high-elevation failure, measured. The deployment supplies the high-elevation ground Section 6.2 identifies as missing. Partitioning all 136 device-hours by which side of the root split each window falls on (Figure 10), ground above the split produced *zero* activations in 63.6 h, while ground below it activated for 66.9 of 67.2 hours. One session scored 1,450 of 1,450 windows positive across 12.1 unbroken hours while the participant drove. Figure 11 shows the shape: the detector is active for essentially the entire Yellowstone stay on both devices, and the two flights are the two narrow bands. At Bozeman, 1,360 m elevation, gate pressure was 105 hPa below the split and every window in the hour before pushback was active. The corpus contains no comparable ground window: its ground minimum sits 237 hPa above the lowest ground pressure observed here.

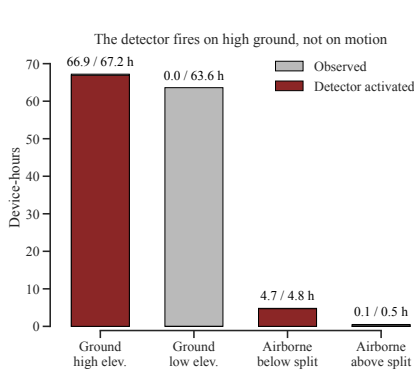


Figure 10: Device-hours observed (gray) against hours active (red), by phase and side of the root split. Ground below the split is active for 66.9 of 67.2 h.

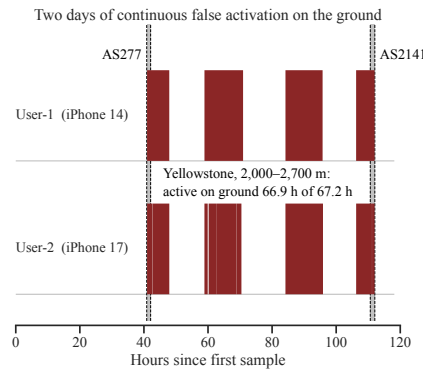


Figure 11: Detector state across the trip, both devices. Shaded bands mark the two flights; almost all active time lies outside them, on the ground at 2,000–2,700 m.

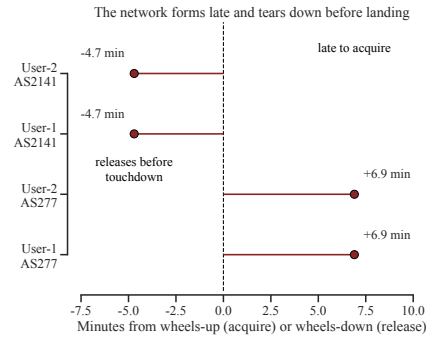


Figure 12: Activation relative to wheels-up, release relative to wheels-down. Both devices acquire at +6.9 min and release 4.7 min *before* touchdown.

Low-elevation deployment had zero activations over 2.65 ground-days, consistent with but far too small to validate the predicted 0.6 per day.

Latency, and a release-side failure the corpus cannot show. Both devices acquired at +6.9 min after wheels-up (Figure 12), close to the corpus median. Only 120 s is debounce; cabin pressurization accounts for the preceding 4.9 minutes. That first window fires on L8, and `pressure_mean` does not cross the split until +6.4 min, so an L1-only rule would have activated 1.5 minutes later. Both devices then *released* 4.7 minutes before wheels-down as cabin pressure recrossed the split. The identical behavior across iPhone generations indicates a rule-level failure that window-level scoring cannot express: the gate closes while still airborne.

Cross-device reproducibility. Across 7,311 jointly scored windows, the devices agree on 99.69% (Cohen’s $\kappa = 0.994$), disagreeing on 23. With route held fixed, the rule is highly reproducible and reproducibly wrong about elevation, complementing Section 6.3’s matched 0.127 gap.

Which leaves fire, conditioned on phase. Conditioned on phase, L8’s 16 firings are all airborne across 136 hours of mixed driving and flight, while L7’s two ground firings are absorbed by debounce (Appendix D).

A contract-violating signal that does separate. Sustained pressure *change* separates the field data where level does not: climb and descent segments clear ground by 30.2 hPa, including mountain driving, whereas peak rate alone does not separate at all (Appendix D). This is the pressure-trajectory signal the Conclusion names. The field data quantifies its margin while confirming it is stateful and therefore outside the deployment contract, so it measures that contract’s cost rather than repairing the detector within it.

Scope. The study establishes that these failures exist, what drives them, and how severe they are; it does not establish how often they occur in the field. One participant pair, one itinerary, four device-legs and 5.3 airborne device-hours. The elevation result is sustained over roughly 131 ground device-hours; every flight-side number is a case study car-

rying no interval. The two devices shared weather, elevation and itinerary, so their errors are correlated and per-device rates cannot be composed under independence. Table 2 collects the comparison and Appendix D the detail behind it.

7 Conclusion

An overlap is not a coverage gap. A smaller corpus made barometer failures look like missing negative coverage. Later flight data populated the same pressure region and exposed true overlap: threshold movement trades false positives for false negatives. Corpus size alone is therefore insufficient; which parts of the overlap are sampled determines which failures are visible.

What the analysis changed. The analysis changes five deployment decisions. The overlap is a documented limitation, not a negative-data defect. Latency has separate scheduling and overlap causes, with overlap more often causing non-detection. Thresholding plus debounce reduces false activations from 33.8 to 0.6 per ground-day, or about 1.8 under nested selection, at 450 s median latency over 15 of 19 activated recordings. Fusion is rejected because added sensors impose permission, power, and availability costs. Most importantly, field deployment turns high elevation from a scope caveat into a shipping blocker requiring a safeguard.

What would help. Separating cases where pressure level overlaps requires new information. In the field, climb and descent segments separated from ground by 30.2 hPa in net pressure change, but no trajectory classifier was evaluated and any such feature violates stateless contract. Contract-preserving options are an abstention region or another modality informative inside the band. More mountain-driving data should move the threshold, not remove overlap.

Limitations. The corpus has eleven both-class trips across seven iPhone models, with limited multi-passenger coverage. The field study is one participant pair on one itinerary; its ground result is sustained, while flight-side results are case studies and device errors are correlated. Deployment event rates rest on very few events and indicate order of magnitude. Appendix F details remaining limitations.

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A Corpus and Feature Details

A.1 Confounder Survivability

Figure 13 gives the fraction of the barometer’s false-positive runs that survive a $k=5$ debounce, by scenario. Driving errors clear entirely, because no run exceeds four windows; lift errors reach eight windows and one third survive, and errors at elevation survive any finite debounce while the device remains elevated. Per-window false-positive rate therefore does not predict deployability, and Table 3 should be read alongside this figure rather than alone.

A.2 Corpus Composition

Table 5 gives the per-trip composition of the corpus, and Table 4 the settled cabin pressure per trip behind Section 4.2’s 753–838 hPa range.

A.3 Feature Definitions

Table 6 lists each sensor’s within-window statistics. Figure 14 plots `pressure_mean` as raw window fractions. Because ground outnumbers flight about five to one inside the overlap band, a shared count axis visually suppresses the in-flight tail that drives the failure analysis.

A.4 Window Length

Changing the window from 5 to 60 seconds moves barometer pooled AP by only 0.0017, far below the 0.0492 resolution floor (Figure 15); the accuracy effect is therefore unresolved. Window length does not resolve false alarms either. Windows fall from 1,267 per day at 5 s to 96 at 60 s, but that is the denominator: the rate moves only 0.073 to 0.067, and false-positive time 106 to 96 minutes per day.

B Detection and Failure Detail

Section E.5 explains the diagnostics that govern interpretation of everything below.

B.1 Detection ranking and hard windows

Figure 16 plots the first two columns of Table 1: the top four sensors separate, and below them the ordering is capacity-bound rather than informative. Figures 17 and 18 give the two views behind Section 6.2’s coverage analysis. Recall on ordinary in-flight windows is 0.975 [0.955, 0.987] under the tree and 0.964 under logistic regression, against 0.158 and 0.061 on approach-equalized windows: the collapse is a property of the population, not of the model class. Defining hard windows by cabin pressure regime rather than by proximity to a transition isolates the failure far more sharply: normalized AP -0.857 [$-1.325, -0.133$] for the ground-pressure zone against 0.322 [0.175, 0.478] for windows within four minutes of a transition.

B.2 Repairs: temporal state and debouncing

Figures 19 and 20 detail Repair 2. The state effect is small, changes sign between model classes, and below eight training trips random subsets straddle zero, so its direction is not resolvable at this corpus size. Figure 21 details Repair 3: raising k removes distinct activations far faster than it removes false *windows*, because overlap errors arrive in runs rather than in isolation.

C Additional Results

C.1 Availability-Aware Scoring

Table 7 scores every modality on the barometer’s out-of-fold window grid, assigning $p = 0$ when a sensor is silent. Motion-coprocessor sensors are always present and unchanged; Location and microphone lose the most, matching deployment history.

C.2 Fusion Detail

To test whether other sensors fill concentrated barometer failures, and what barometer-free devices can do, we use late fusion: each sensor emits a window probability and a logistic meta-classifier combines them. Nested cross-validation supplies meta-features from inner leave-one-trip-out predictions, preventing base-model exposure to the outer test trip. Sensors are outer-joined on the barometer’s decision grid: a window a sensor never observed enters as $p = 0.5$ with a presence indicator rather than being dropped, so the comparison is not complete-case.

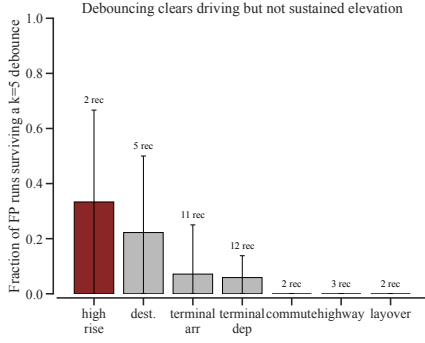


Figure 13: Fraction of barometer’s false-positive runs surviving a $k=5$ debounce, per confounder scenario. Driving clears; lift and elevation errors survive.

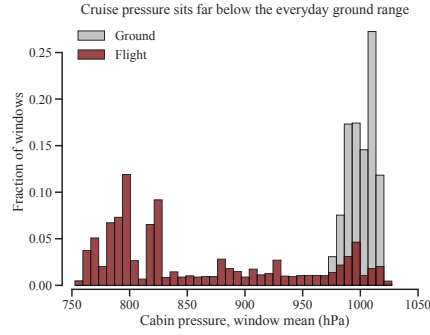


Figure 14: Cabin pressure on the ground against in-flight, as raw window fractions. Ground outnumbers flight inside the band, which Figure 4 rescales away.

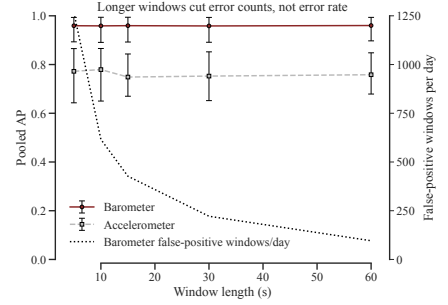


Figure 15: Pooled AP against window length, with false-positive windows per day on the right axis. The AP span is far inside the ± 0.049 resolution floor.

Table 3: Confounder hold-out (P2): fraction of ground-only windows predicted in-flight, i.e. the false-positive rate. **The five columns shown are those discussed in the text; worst is the maximum over all eight scenarios**, so it can exceed every value shown.

Sensor	High-rise	Highway	Commute	Term. arr	Term. dep	Worst
Barometer	0.047	0.027	0.072	0.146	0.014	0.146
Location	0.000	0.000	0.048	0.238	0.049	0.238
Microphone	0.037	—	0.347	0.239	0.144	0.347
Brightness	0.005	—	0.000	0.108	0.102	0.562
Compass	0.315	0.380	0.260	0.375	0.578	0.618
Accelerometer	0.140	0.295	0.263	0.352	0.450	0.600
Magnetometer	0.720	0.060	0.160	0.335	0.562	0.720
Battery	0.042	—	0.000	0.830	0.244	0.830
Orientation	0.203	0.820	0.370	0.379	0.606	0.820
Gyroscope	0.308	0.959	0.622	0.602	0.625	0.959
Gravity	0.284	0.975	0.645	0.767	0.691	1.000
Network	0.182	0.000	0.210	0.087	0.064	1.000
Pedometer	0.667	1.000	0.767	0.835	0.942	1.000

All-sensor fusion reaches pooled AP 0.984 [0.954, 0.998] versus barometer’s 0.958 [0.895, 0.993] (Figure 22). The gain is useful only where the added sensors’ power, permission, and availability costs are acceptable (Section 6.1).

Without barometer, pooled AP is 0.926 [0.856, 0.977], while greedy forward selection peaks at 0.939. That peak is optimistic because selection and scoring use the same folds. Excluding each held-out trip’s score from set selection lowers mean held-out AP to 0.901, a $+0.038$ selection bias. Location and microphone appear in all eleven selected sets, but the third sensor alternates between accelerometer and gyroscope, and held-out AP ranges from 0.612 to 0.999. The instability precludes a fixed recommendation. Barometer-free detection is feasible as a fallback but is less accurate and has more false alarms.

C.3 Cross-Device Transfer

On two trips, two passengers recorded the same flight simultaneously on different devices, controlling route, weather, and aircraft. Across four shared sensors, the mean device gap is 0.070; per-sensor gaps range 0.023–0.127 on two cap-

tures each (Figure 9). Barometer’s 0.127 is inflated by one paired recording entirely in near-field-pressure flight, scoring 0.236 versus its partner’s 0.816. The cleaner finding is a reproducible level offset. Across three simultaneous pairings with 290, 210, and 173 shared windows, iPhone 16 reads 0.531, 0.497, and 0.551 hPa above iPhone 13 Pro Max, with within-pair standard deviations 0.055, 0.085, and 0.049. Pressure changes agree while absolute levels differ, so an absolute-pressure rule inherits device bias. Leave-one-device-out spans 0.081–0.970 but is route-confounded (Figure 23).

C.4 Calibration and Operating Point

Calibration and operating point. Because false activations cost more than misses, threshold 0.5 is unsuitable. Barometer probabilities are reasonably calibrated (Figure 24), and a 10:1 false-positive cost shifts the operating point substantially (Figure 8). We report a range because trip-level variation makes a single optimum overprecise. The exact cost ratio is not load-bearing: $\theta = 0.843$ remains optimal from 2:1 through 20:1, versus 0.826 at 1:1 and 0.963 at 50:1, where

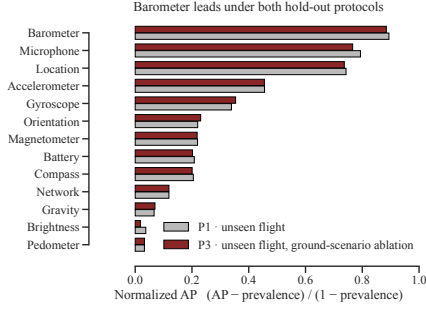


Figure 16: Normalized AP per sensor under P1 beside P3, the graphical form of Table 1. The top four separate; below them the ranking is capacity-bound.

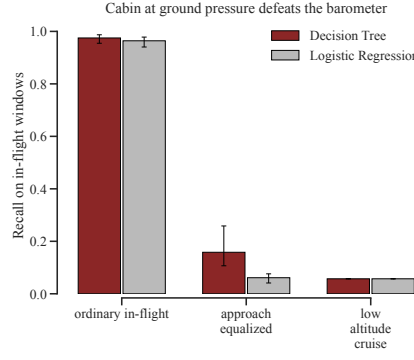


Figure 17: Barometer recall on three in-flight populations, both model classes. Recall collapses off ordinary flight; the low-cruise arm rests on two recordings.

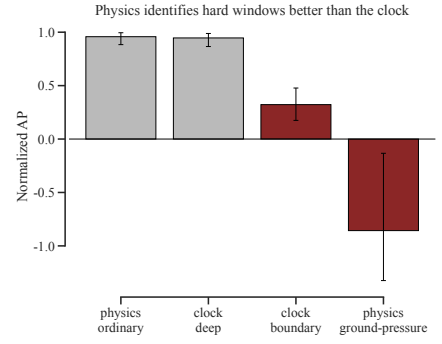


Figure 18: Hard windows defined by proximity to a transition (clock) against cabin pressure regime (physics), in normalized AP. Physics separates far better.

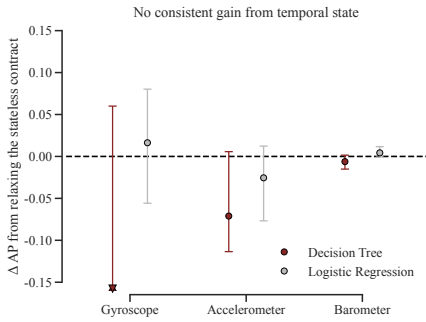


Figure 19: Δ AP from relaxing the stateless contract, per sensor, under both model classes. The barometer’s sign changes between them.

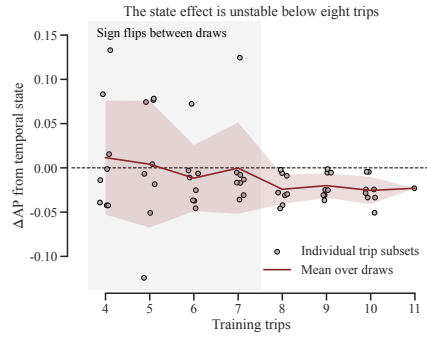


Figure 20: Δ AP from temporal state against the number of training trips, eight random subsets each. Below eight trips the draws straddle zero.

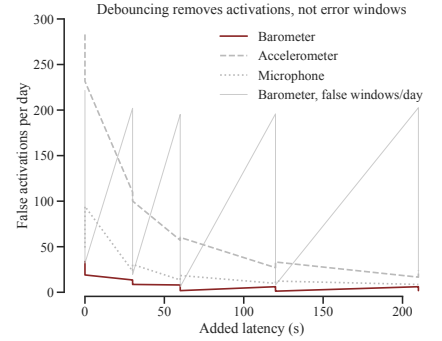


Figure 21: False activations per day against added latency, at untuned 0.5 threshold. Activation rate falls five-fold while false-window rate barely moves.

0.001 lower false-positive rate costs 0.029 recall. With no ex ante activation budget, we report the achieved 0.6 false activations per ground-day versus 33.8 untuned and leave the final operating point to the product objective.

C.5 The Selected Configuration and Its Behavior

Table 8 restates the selected configuration and reports what it does on the corpus, at the operating point Section 6.3 selects. Section 6.4 tests the latency and false-activation rows prospectively.

C.6 Policy Selection Is Not Free

Table 8’s operating point is selected and scored on the same out-of-fold predictions. Out-of-fold scoring prevents model-fit leakage but not bias from choosing θ and k after seeing held-out labels. We therefore nest policy selection: for each held-out trip, choose θ and k under the same 10:1 criterion using only the other trips’ out-of-fold predictions, then apply the fixed policy to the held-out trip.

Nested selection raises false activations from 0.61 to 1.84 per ground-day and lowers window recall from 0.867 to

0.837. The policy is unstable: across eleven folds, θ takes three values from 0.793 to 0.843, and k takes two.

Both rates, however, rest on few events: one in-sample activation versus three nested over 4,691 ground windows, or 1.63 days. Exact Poisson intervals, $[0.02, 3.42]$ and $[0.38, 5.38]$ per day, overlap heavily, so the optimism is not resolved. The corpus supports only scale and direction: tens of activations per ground-day untuned, roughly one after cost-sensitive thresholding and debounce. We report 0.6 as achieved on this corpus, not an unbiased deployed rate; prospective field data are required.

C.7 Deployment Efficiency Score

Detection accuracy alone does not determine deployment fitness. Table 9 rates power, privacy, background availability, and sampling cost per modality; mapping Low/Full to 1, Medium/Partial to 2, and High/Restricted to 3 gives the illustrative composite

$$\text{efficiency} = \frac{\text{AP}}{\text{power} \times \text{privacy} \times \text{background}}.$$

Sampling cost is shown but excluded because power already captures it for these sensors. The rank product is descrip-

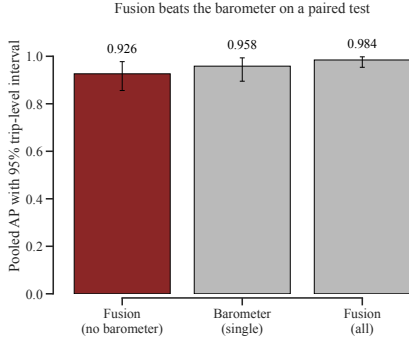


Figure 22: Late fusion against single sensors, trip-level intervals. Fusing all beats barometer on a paired test; barometer-free stack is lower but unresolved.

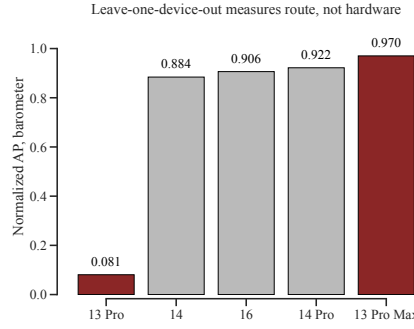


Figure 23: Leave-one-device-out for barometer. *Confounded*: each device recorded different routes, so spread measures route difficulty rather than hardware.

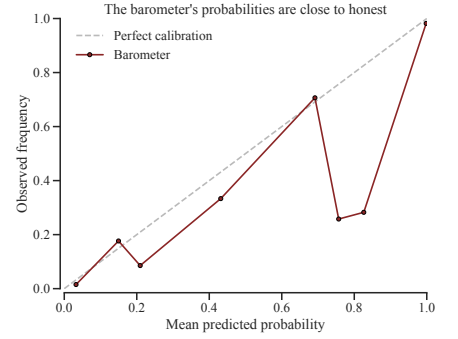


Figure 24: Reliability curves against the perfect-calibration diagonal, for the barometer and the accelerometer. Points below the line are over-confident.

Table 4: The eleven both-class trips forming the leave-one-trip-out folds, ordered by settled cabin pressure (5th percentile of in-flight window means), which spans 85 hPa across trips.

Trip	Cabin (hPa)	Legs	Ground	Flight
T31	752.6	1	320	57
T34	761.4	1	166	484
T12	771.9	2	594	304
T23	772.8	1	1,550	198
T18	781.4	2	512	316
T25	781.5	1	186	198
T30	788.0	1	5	579
T29	797.9	1	17	652
T22	818.0	2	491	230
T20	825.7	2	297	240
T36	838.0	2	553	414

tive, not calibrated, and no conclusion depends on it. Section 6.1’s recommendation rests on the underlying dimensions, where barometer leads in accuracy, power, privacy, and background availability; sampling cost is the only axis on which it is not lowest.

D Field Study Detail

The measurements Section 6.4 defers.

Figure 25 gives per-leg rendezvous availability, the fraction of each flight during which both devices were simultaneously active (90.6% of AS277 and 92.1% of AS2141, counting only the time each active window occupies inside the leg), and Figure 26 the liveness budget behind it: of the 118.2 h common span, both devices scored for 60.9 h and exactly one for 14.5 h, the latter almost entirely an app one participant had not yet started. Figure 27 conditions leaf occupancy on flight phase. L8 fires 16 times and every one is airborne; L7 fires twice and both are on the ground, at sea level under high pressure after landing. Both were isolated single windows and the $k=5$ debounce absorbed them, which

Table 5: Dataset composition, one row per trip. A *trip* is a whole travel episode (every leg of a connection and every device that recorded it) and is the unit held out by P1. Confounder trips are ground-only and are tested through P2.

Trip	Device(s)	Ground	Flight	Role
T12	14	594	304	fold
T18	14	512	316	fold
T20	13 Pro	297	240	fold
T22	13 Pro	491	230	fold
T23	14	1,550	198	fold
T25	14	186	198	fold
T29	13 Pro Max	17	652	fold
T30	13 Pro Max	5	579	fold
T31	14 Pro, 15 Pro	320	57	fold
T34	13 Pro Max, 16	166	484	fold
T36	13 Pro Max, 16	553	414	fold
T39	14 Plus	381	0	confounder
T40	13 Pro	637	0	confounder
T43	14 Plus	48	0	confounder
T01	13 Pro	0	65	train only
Total (39 rec.)	7 models	5,757	3,737	11 folds

is that mechanism working as intended. “Change-keyed implies airborne” therefore holds for L8 and fails for L7, and L7 rests on 11 training samples, so it should be read as unvalidated rather than as evidence that `pressure_range` misfires on the ground.

Figure 28 gives the pressure-change separation by segment role. The eight climb and descent segments span 80.4–224.1 hPa of net change against a ground maximum of 50.2 hPa over 58 segments, a margin of 30.2 hPa; peak rate alone does not separate, at 7.01 against 6.13 hPa/min. Two mid-cruise step changes fall at 24 hPa, below the ground maximum, but they occur inside an already-detected flight and need not separate; pooling them with climb and descent is what makes the margin appear to vanish. Figures 29 and 30 document the field deployment build’s sample-count

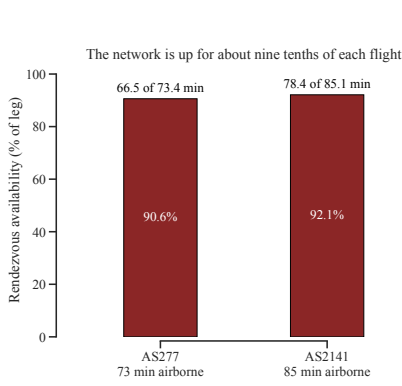


Figure 25: Fraction of each leg with both devices simultaneously active, the condition a flight-scoped peer network requires.

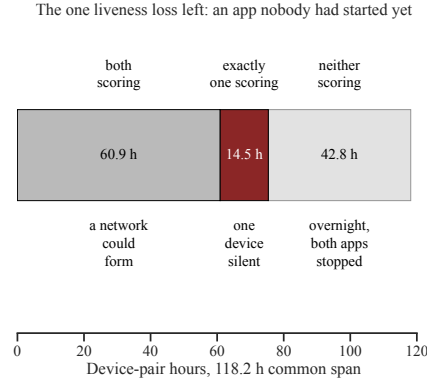


Figure 26: The 118.2 h common span, split by whether a network could form at all. The 14.5 h one-sided residual is almost entirely an app not yet started.

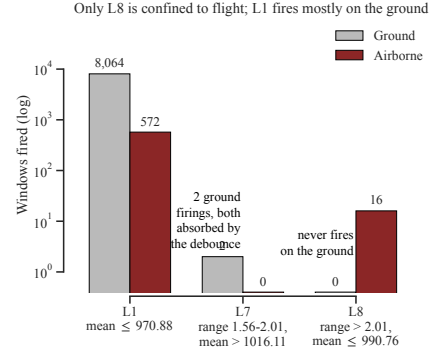


Figure 27: Leaf occupancy conditioned on flight phase, log scale. L8's firings are all airborne; L7's two are both on the ground. Pooled counts hide both facts.

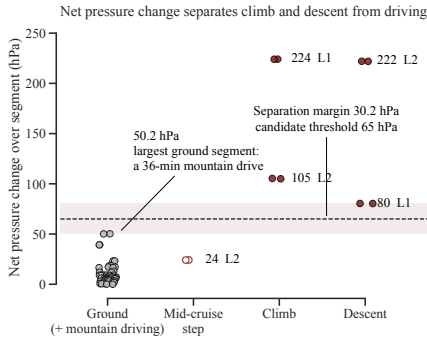


Figure 28: Net pressure change per sustained segment by role. Climb and descent separate from ground with 30.2 hPa margin; mid-cruise need not.

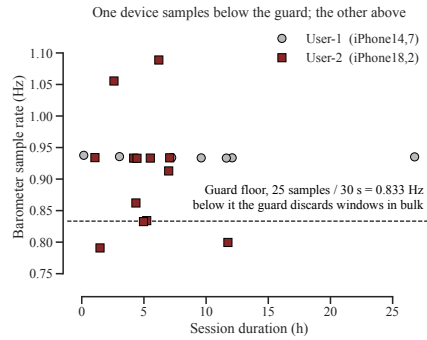


Figure 29: Per-session barometer sampling rate against session length. One device holds 0.933–0.938 Hz; the other dips below the guard (0.791–1.089 Hz).

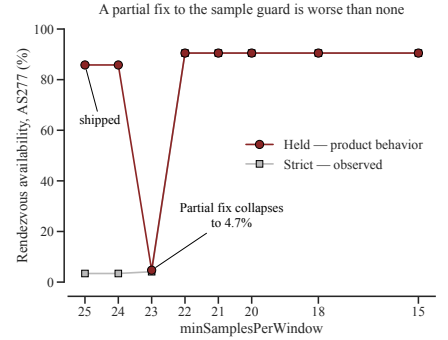


Figure 30: Rendezvous availability on AS277 against the sample-count guard, swept on both devices. A partial correction is worse than none.

guard, the one implementation defect the subsequent deployment corrects: per-session sampling rate on one device ranges 0.791–1.089 Hz, and because the guard is an absolute count rather than a fraction of the expected count, availability responds non-monotonically to it. A partial correction is worse than none, since the device then scores often enough to reset the debounce but not often enough to sustain it.

E Method Details

E.1 Hold-out Protocols and Fold Construction

Table 10 summarizes the protocols. Folds hold out whole travel episodes, not individual legs, so connected legs, layovers, terminal windows, and simultaneous recordings from the same episode cannot cross train-test boundaries. Flight-only recordings train but are not tested; ground-only confounders are tested under P2. Confounder holdout also groups recordings sharing the same condition: all three driving legs are held together because consecutive legs share an air mass, and both high-rise sessions together to prevent memorizing the building.

E.2 Applying the Protocol

Applying the protocol. For a new always-on detector, identify the physical quantity exploited by the strongest signal. Enumerate realistic negatives that move that quantity similarly without the target event, and record them at deployment-scale duration with the full sensor suite. Hold each scenario out as its own fold and report fold-level performance under the same metric as ordinary folds. If performance degrades, add that condition to training and rerun the same folds while measuring the cost on the population not repaired. This distinguishes a coverage gap from a representational limit, as in Section 6.2.

E.3 Metric Definitions

Resolution floor. With eleven trips, effects below some magnitude cannot be resolved. We use a *resolution floor* as a screening heuristic, defined as the half-width of the trip-level interval for the quantity of interest and call smaller differences unresolved, not absent. For barometer absolute AP, the floor is ± 0.049 . It is a single-arm half-width, so where two arms share folds we report a paired trip-level interval

Table 6: Per-sensor within-window features. Every statistic is computed inside a single 30-second window, with no dependence on earlier windows, which is the feature half of the stateless contract.

Sensor	Within-window features
Barometer	pressure mean/std/range
Accelerometer	magnitude & 1g-deviation stats; per-axis std
Gyroscope	angular-velocity mean/std/max; per-axis std
Gravity	per-axis mean/std; magnitude mean/std
Magnetometer	field magnitude mean/std; per-axis mean
Orientation	roll/pitch/yaw mean/std
Compass	magnetic bearing mean/std
Location	GPS validity; speed mean/max; altitude; h. acc.
Microphone	dBFS mean/std/max
Battery	level mean; charging fraction
Brightness	brightness mean/std
Network	connected/WiFi/cellular fractions
Pedometer	step rate (steps/s within window)

instead, which resolves smaller effects than the floor admits. *Model class as a control.* Sensor-ranking and failure-mechanism claims use both a decision tree and logistic regression on identical folds; deployment choices (θ , debounce, Table 8) belong to the tree. The models differ structurally: a tree can isolate bounded pressure intervals, whereas logistic regression fits a monotone boundary. Effects seen in only one are therefore model-dependent. Random forest and depth-6 gradient boosting instead form a *capacity ceiling* (Figure 1); because they share tree inductive bias, they are not independent controls.

E.4 Leave- k -Out Robustness

Robustness. Across all $741 = \binom{39}{2}$ leave-two-recording subsets, barometer fold-mean raw AP under P1 stays within $[0.932, 0.940]$ except when recordings 20 and 21 are both removed, yielding 0.898. This probes sensitivity to captures, not whole travel episodes. No leave-one or leave-two subset changes any reported effect’s sign. Temporal-state Δ AP versus corpus size (Figure 20) likewise shows sample-size sensitivity: below eight trips signs vary across subsets; above eight the spread contracts roughly fourfold.

E.5 Model-Class Diagnostics

The following figures are diagnostic apparatus, not separate flight-detection findings. Figure 31 compares confounder false-positive rates under the decision tree against logistic regression; points off the diagonal describe the model rather than the sensor, which is why Section 6.3 treats failure direction as robust and magnitude as model-dependent.

Figure 32 diagnoses the earlier interpretation from Section 6.2. Under logistic regression the effect never exceeds 0.0003; under the tree it stays at or below 0.0016 through depth three and reaches 0.0254 at depth four. None of twelve capacity $\times$ weighting intervals excludes zero.

Table 7: Coverage of the decision grid, and normalized AP scored where the sensor reads versus scored on the whole grid.

Sensor	coverage (%)		norm. AP	
	flight	ground	observed	fail-safe
Barometer	100.0	100.0	0.926	0.926
Accelerometer	100.0	100.0	0.559	0.559
Gyroscope	100.0	100.0	0.429	0.429
Battery	85.7	76.4	0.368	0.340
Location	36.1	72.6	0.944	0.336
Network	98.5	93.1	0.309	0.321
Microphone	31.0	50.1	0.868	0.261
Magnetometer	100.0	100.0	0.259	0.259
Orientation	100.0	100.0	0.218	0.218
Brightness	85.7	76.4	0.130	0.147
Compass	100.0	100.0	0.121	0.121
Gravity	100.0	100.0	0.100	0.100
Pedometer	100.0	100.0	-0.026	-0.026

Figure 1 tests capacity, not inductive-bias robustness: ensembles can extract more signal but share the tree’s structure (Section 5).

Figure 33 motivates the resolution floor by showing how resampling unit changes interval width (Section 4.3).

E.6 Representation Ceiling

Section 6.2 attributes the failure to one-window absolute-pressure observability. The capacity ceiling varies models over three summary features; here we instead ask whether a richer representation of the same permitted window separates the hard cases.

Each 30-second barometer window contains a median 28 samples at 0.9 Hz. We expand the deployed three statistics to 56 features: 16-point resampled sequences at absolute and mean-centered levels, seven quantiles with IQR, skew, kurtosis, slope, curvature, first-difference statistics, and lag-1 autocorrelation. No cross-window state is used. Random forest and gradient boosting match the capacity-ceiling estimators, isolating representation as the changed factor. Models use identical P1 folds.

Table 11 confirms that the hard zone remains unresolved: normalized AP ranges from -0.678 to -1.198 and recall from 0.060 to 0.132 across configurations. The richest model recalls exactly the same fraction as the deployed tree (0.132), and representation differences lie within the existing $[-1.325, -0.133]$ interval.

Richer features improve the broader 960–1010 hPa band, raising normalized AP from 0.366 to 0.512, but do not repair the overlap. That band contains 3,289 windows at roughly 16% prevalence and is mostly ordinary ground; the hard zone contains 279 windows at 95% prevalence. Richer one-window features therefore improve the easy majority while leaving the physically overlapping population unchanged.

F Limitations in Detail

Sensor support varies across the corpus: microphone uses five trips against eleven for the motion sensors, and Net-

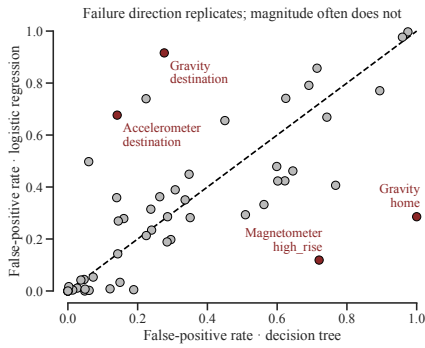


Figure 31: False-positive rate per (sensor, scenario) under decision tree against logistic regression. Points off the diagonal describe the model, not the sensor.

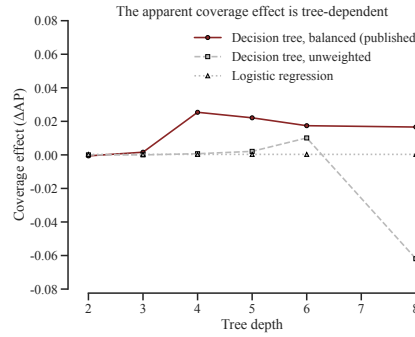


Figure 32: The coverage effect against `max_depth`, both weightings and both model classes. No cell has an interval excluding zero.

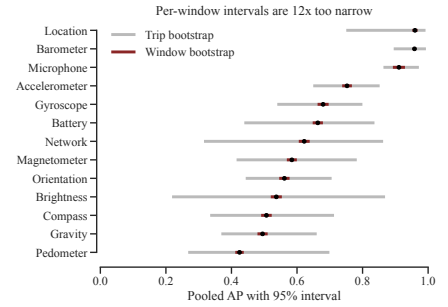


Figure 33: The same predictions resampled two ways for 13 sensors: window bootstrap against trip bootstrap. Window interval is a median $12\times$ too narrow.

Table 8: The selected configuration and its corpus-measured behavior. Fixed before the deployment of Section 6.4 and never refit. Latency is from wheels-up, with latency from the detector’s first observed window in parentheses. [†]Median and P90 are over the 15 recordings that are detected; the *within-m* rows are over all 19. [‡]Selected and scored on the same out-of-fold predictions, over 1.63 ground-days; nested selection gives 1.84 (Appendix C.6).

Configuration	
Sensor	Barometer only (3 features)
Window	30 s, no cross-window <i>features</i>
Classifier	Depth-4 decision tree
Threshold θ	0.843
Debounce	$k_{\text{on}} = k_{\text{off}} = 5$
Cold start	Closed (fail-safe)
Behavior	
False activations	0.6 per ground-day [‡]
Window recall	0.867
Recordings detected	15 / 19
Median latency [†]	450 s (330 s)
P90 latency	7,140 s (984 s)
within 2 min	0 / 19 (3 / 19)
within 5 min	4 / 19 (7 / 19)
within 10 min	10 / 19 (13 / 19)
within 30 min	11 / 19 (14 / 19)

work is weak because nearly all flight recordings use airplane mode, so cross-sensor comparisons at the bottom of Table 1 rest on different amounts of evidence. Low-altitude cruise has only two recordings, so we report no interval for it; the full overlap band spans fifteen. Each confounder scenario comes from one session or a small set on one device, which is enough to establish that a mechanism produces flight-like signal but not to estimate how often it does.

Greedy sensor selection hides the held-out trip’s score but not its influence on base-learner fitting, so it diagnoses selection stability, not performance; the all-sensor stacker (Appendix C.2) is fully nested. Thirty-nine secondary sensor $\times$ protocol cells accompany the primary end-

Table 9: Deployment characterization. Power, privacy and background availability are design facts about iOS rated 1–3; sample rates are measured. *Background* records whether the sensor reads whenever the app receives execution time, not whether iOS grants that time; scheduling delays are visible in the latency tail. Efficiency is pooled AP over the cost product, and is illustrative rather than an analysis.

Sensor	Power	Priv.	Back.	Hz	AP	cost	eff.
Barometer	Low	Low	Full	0.9	0.958	1	0.958
Battery	Low	Low	Full	1.0	0.663	1	0.663
Network	Low	Low	Full	0.5	0.622	1	0.622
Brightness	Low	Low	Full	1.0	0.537	1	0.537
Accelerometer	Low	Med	Full	99.7	0.753	2	0.376
Pedometer	Low	Med	Full	0.1	0.425	2	0.212
Orientation	Low	Med	Partial	99.7	0.561	4	0.140
Gravity	Low	Med	Partial	99.7	0.495	4	0.124
Gyroscope	Med	Med	Partial	99.7	0.680	8	0.085
Magnetometer	Med	Med	Partial	99.7	0.584	8	0.073
Compass	Med	Med	Partial	99.7	0.507	8	0.063
Location	High	High	Restricted	0.5	0.960	27	0.036
Microphone	High	High	Restricted	9.6	0.911	27	0.034

point; about two would clear by chance, so we treat individual secondary cells as descriptive.

Finally, the corpus itself never samples sub-split ground pressure. Its minimum ground reading is 970.9 hPa against a learned split of 970.88, so every ground window falls on the ground side by construction, a coincidence that made the split look safer than Section 6.4 shows it to be.

Table 10: The three hold-out protocols. P3 is the primary endpoint; we report P1 alongside it because P1 is what the leave-one-group-out results common in this literature correspond to.

	Held out	Question
P1	a trip	unseen flight
P2	a ground scenario	unseen deployment condition
P3	trip; scenario dropped from training	unseen flight, ground ablation

Table 11: A richer read of the same window. The ground-pressure zone is the population of Section 6.2: 279 windows at 0.95 prevalence. Norm. AP is normalized against each population’s own prevalence.

Model (features)	overall	ground-pressure zone	
	norm. AP	norm. AP	recall
Tree, depth 4 (3)	0.926	−0.857	0.132
Logistic regression (3)	0.955	−0.956	0.060
Random forest, 400 (3)	0.950	−1.198	0.128
Random forest, 400 (56)	0.950	−0.906	0.124
Gradient boosting (56)	0.946	−0.678	0.132